

helix_proxy — VPN-LAN

Service Access Feature Status

Revision: 8 **Last modified:** 2026-07-02T09:39:19Z **Status:** Feature COMPLETE (Phases 0-12 on main). Anti-bluff bridge scaffold + preflight doctor + operator guide + Challenge runner + **Phase-1 SSRF carve-out teeth** + **Phase-12 bidirectional ingress-allowlist teeth** + the §11.4.135 autonomous battery (stress+chaos, benchmark+memory, concurrency+load, DNS-rebinding gap) are PROVEN NOW (autonomously, no live VPN, no podman) and wired GREEN into the standing suite. **NEW (§11.4.52 hermetic promotions):** the Chromecast-eureka, FTP-passive, WebDAV-PROPFIND, and email (SMTPS/IMAPS/POP3S) protocol legs now ALSO run **AUTONOMOUSLY** — their client-side protocol logic proven over a real rootless kernel-WireGuard tunnel between two unprivileged netns against a pure-stdlib peer (zero installs, no operator/podman/VPN), each with golden-bad teeth + a not-stale self-fetch + 3/3 determinism + independent GO; **all four protocol legs — Chromecast-eureka, FTP-passive, WebDAV-PROPFIND, Email — plus the H0 WG substrate are wired into the standing suite as §11.4.135 regression guards (test_vpn_lan_hermetic in tests/run-tests.sh, SKIP-aware)** (the email leg was wired in b66b2fa this session, closing the §11.4.6 gap the §11.4.169 ledger audit had caught — independent review a5bc6266 confirmed all 5 rows PASS in the real loop); the **live** Mullvad round-trip stays operator-gated (these COMPLEMENT, never replace, §11.4.6). Every live protocol round-trip (SMB/NFS/FTP/SFTP/WebDAV/email/Cast/DIAL/ADB) is AUTHORED with real byte-integrity / sink-side evidence gating **plus a Phase-12 both-way reverse-leg assertion** (NFS NLM/NSM callback, FTP active, Cast callback, adb reverse, email N/A §11.4.6) and honestly SKIPS while the sword bridge is down (§11.4.3) — never a fake PASS; Miracast is Won't-fix structurally-impossible (§11.4.112). Operator-gated: the sword bridge (secrets + Mullvad + root/sudo) → unlocks all live round-trips BOTH directions; the multicast-reflector remote deployment; any real-device flash. **Data-plane env-block (operator-actionable, NOT a code defect):** host rootless-podman aardvark-dns cannot bind netavark :53 → :53128/:51080 don't serve + ./start fails; the standing suite honestly reports this (3 env FAILs) rather than hanging. **Authority:** Inherits constitution/Constitution.md per §11.4.35. This is the §11.4.153 per-feature Status set (companion §11.4.56 two-audience Status_Summary.md) for the VPN-LAN service-access feature workstream (feature/vpn-aware-dynamic-

routing, §11.4.167). Every row is reconciled against real committed code + tests + design docs; no metadata-only PASS and no fabricated video-confirmation (§11.4 / §11.4.1 / §11.4.6). **Companion / cross-refs:** integration status [../design/vpn_lan_access/Status.md](#) · plan [../design/vpn_lan_access/PLAN.md](#) · Miracast verdict [../design/vpn_lan_access/miracast_verdict.md](#) · reflector design [../design/vpn_lan_access/reflector_design.md](#) · operator guide [../guides/vpn_lan_bridge_setup.md](#). **Repo anchor:** branch main (all work merged to + performed on main per operator directive). Landing commits: bridge scaffold d781002 + standing-suite wiring 5c28f56; Phase 2/3 file protocols 182e80a; Phase 4 email 2f31460; Phase 5 reflector d0b42df; Phase 6/7 Cast+ADB 65043ce; Phase 8 Miracast 12faf12; Phase 9 guide a5e5616; Phase 11 Challenge runner + HelixQA bank 89f73b7; Phase 1 SSRF teeth 1d87b3a; Phase 10 containerization 1911d5e; Phase 12 bidirectional 2ed0fed + both-way protocol assertions cc8b620; stress+chaos 97d9733; bench+memory + dns-rebinding 4e2810c; concurrency+load 7e73b08; §11.4.135 suite guards 6d8893f; **hermetic H0 substrate + H2 Chromecast-eureka promotion 18a21bd; H2.ftp + H2.webdav promotions 3b98d02** (§J, §11.4.52).

Operator-blocked / operator-gated items (read first — §11.4.45 O(1) surface)

Every autonomous claim below is either PROVEN NOW against a committed file/evidence artefact, or an honest SKIP that flips to a live PASS the moment the operator supplies the input named here. No live round-trip is marked PASS without captured evidence (§11.4.6 / §11.4.69). **Note (§J):** the FTP-passive, WebDAV-PROPFIND, and Chromecast-eureka legs' *client-side protocol logic* is ALREADY proven AUTONOMOUSLY over a hermetic WireGuard tunnel (no operator/VPN needed) — only the *live Mullvad-topology* round-trip below stays operator-gated.

Item	Why blocked / gated	Unblock condition
Live protocol round-trips (SMB / NFS / FTP / FTPS / SFTP / WebDAV / IMAP / SMTP-submission /	Need the live sword bridge — secrets + Mullvad WireGuard + root/sudo on the bridge side (§11.4.21 / §11.4.10). The scaffold proves the honest-SKIP path + the anti-bluff gate NOW; the live byte/sha256/mailbox/serial evidence exists only when the bridge is genuinely up. Bridge-down ⇒ honest SKIP	Operator copies <code>.env.example</code> → <code>.env</code> , points the 6-var contract at <code>sword_toolkit</code> , brings the VPN up; each test then emits real captured round-trip evidence under <code>qa-results/</code>

Item	Why blocked / gated	Unblock condition
POP3S / Cast / DIAL / ADB)	(network_unreachable_external), never a fake PASS (§11.4.3 / §11.4.69).	vpn_lan/phase<N>/ <ts>/.
Multicast discovery reflector — remote deployment	Deploying an Avahi mDNS / SSDP-relay reflector changes a remote host (§11.4.122) — requires an interactive keep/ deploy decision. The reflector <i>design</i> + the client-side enumeration <i>test</i> are committed and autonomous; only the remote deploy is gated.	Operator authorizes the remote-side reflector deployment (Phase 10 boots it on-demand via the containers submodule, §11.4.76).
ADB flash over VPN (usbip USB- over-IP)	Flashing a real device is high- blast-radius (§11.4.133 target-hardware-safety) and USB-level, not network- routable (fastboot is USB- bound — recon 4 FACT). The test NEVER flashes; it records the honest boundary and SKIPs operator_attended.	Operator authorizes the specific device flash via a remote host with the device attached over usbip.
HelixQA vpn_lan.yaml autonomous session	The helixqa binary cannot be built in this checkout — its go.mod replaces 6 un-vendored own-org sibling modules (§11.4.3). The bank is authored; no autonomous HelixQA session has been run and none is claimed. The runnable proof NOW is the Challenge runner.	Vendor the 6 sibling modules under submodules/, then drive the bank via an autonomous HelixQA session.

Feature status table (§11.4.153)

Legend. *Wiring* = is the feature genuinely reachable/usable by an end user of the deployed helix_proxy through the env-var bridge (§11.4.108)? *Validation* ∈ {PASS, FAIL, SKIP, PENDING_FORENSICS, OPERATOR-BLOCKED}. A **PASS** cites a committed captured artefact or a self-proving standing-suite gate. **SKIP** = honest topology/precondition skip (bridge down / reflector not deployed / server absent — §11.4.3), never folded into a PASS. *Video-confirmation* is **N/A** for every row: helix_proxy VPN-LAN is a headless network feature with no rendered UI — its §11.4.107/§11.4.69 proof is captured exit codes + byte-integrity sha256 + sink-side protocol responses (SMB/NFS/mailbox/eureka JSON/

device serial), not a rendered video (§11.4.153(3) autonomous-infeasible ⇒ honest non-video evidence). Live-path rows additionally carry no autonomous video because the live round-trip needs the operator-gated sword bridge.

A. Bridge foundation & anti-bluff scaffold — PROVEN NOW (autonomous, no live VPN)

Component	Feature	Category	Implementation
env-var bridge contract	6-var decoupled bridge contract (HELIX_SWORD_DIR / _CONNECT / _DISCONNECT / _HEALTH / _SUBNET / _HOST); bridge_load / bridge_up / bridge_require primitives (rc 2 = OPERATOR-BLOCKED, rc 3 = misconfigured)	foundation / decoupling (§11.4.28)	.env.example (tracked, no secrets); tests/ lib/ sword_bridge.sh
svord-doctor preflight	Deterministic 3-verdict preflight — BRIDGE: UP (exit 0) / SKIP:network_unreachable_external (exit 2) / MISCONFIGURED:<reason> (exit 3): env resolvable? hooks executable? HELIX_BRIDGE_HEALTH exit 0? remote host reachable? Never fakes UP	preflight / anti-bluff gate	scripts/ svord_doctor.sh; companion docs/ scripts/ svord_doctor.md (§11.4.18)

B. File shares — AUTHORED, live round-trip operator-gated

Component	Feature	Category	Implementation	Wiring (reachal
SMB / CIFS / NMB	SMB/CIFS write→read-back→sha256 round-trip over the L3-routed VPN share (smbclient, mount.cifs fallback); NMB name-resolution by 10.x unicast	file- share / functional	tests/vpn_lan/ smb_nfs_roundtrip.sh (SMB slice)	Gated — reachabl the bridg + HELIX_VPN set; rout L3, NOT SOCKS5

Component	Feature	Category	Implementation	Wiring (reachal
	(multicast NMB not routed)			
NFS	NFS write→read-back→sha256 round-trip over the VPN export (pre-mounted HELIX_VPN_NFS_MOUNTED, or root temp-mount HELIX_VPN_NFS_EXPORT); temp mount unmounted on every exit (§11.4.14)	file-share / functional	tests/vpn_lan/smb_nfs_roundtrip.sh (NFS slice)	Gated — reachable bridge up export configure routes ov

C. File transfer — AUTHORED, live round-trip operator-gated

Component	Feature	Category	Implementation	Wiring (reachable?)
FTP / FTPS	FTP passive directory-list + fetch over routed :21 + pinned PASV range (PASV must advertise the 10.x addr); FTPS explicit (AUTH TLS) / implicit (:990)	file-transfer / functional	tests/vpn_lan/ftp_sftp_webdav.sh (FTP slice)	Gated — reachable once bridge up + HELIX_VPN_FTP_URL set
SFTP				

Component	Feature	Category	Implementation	Wiring (reachable?)
WebDAV (via existing Squid)	SFTP single-connection byte round-trip over routed :22 (key-based, BatchMode — fully autonomous, §11.4.98); sha256 match required	file-transfer / functional	tests/vpn_lan/ftp_sftp_webdav.sh (SFTP slice)	Gated — reachable once bridge up + HELIX_VPN_SFTP_HOST set
	WebDAV PROPFIND through the EXISTING Squid (no new component) — expect 207 Multi-Status + non-empty XML body; a real non-207 HTTP response ⇒ FAIL (fail-closed §11.4.68)	file-transfer / functional	tests/vpn_lan/ftp_sftp_webdav.sh (WebDAV slice)	Gated — reachable once bridge up + HELIX_VPN_WEBDAV_URL set; traverses HELIX_SQUID_PROXY (default :53128)

D. Email — AUTHORED, live round-trip operator-gated

Component	Feature	Category	Implementation	Wiring (reachable?)
IMAP / IMAPS	IMAPS LOGIN + LIST "" "*" over implicit-TLS :993 (RFC 8314) — assert a real	email / functional	tests/vpn_lan/email_roundtrip.sh (T4.1)	Gated — reachable once bridge up + HELIX_MAIL_

Component	Feature	Category	Implementation	Wiring (reachability)
	mailbox * LIST listing returns; a1 NO/BAD ⇒ FAIL			PASS set; cr via in-proc printf stdin never argv logged (§11.4.10)
SMTP submission	Authenticated submission send over :465 implicit / :587 STARTTLS — a self-addressed round-trip probe token must be 2xx/250-accepted	email / functional	tests/vpn_lan/email_roundtrip.sh (T4.2)	Gated — reachable bridge up creds set
POP3S	POP3S retrieve over implicit-TLS :995 — retrieve the just-sent T4.2 message and verify its unique token (a true send→retrieve round-trip); token not returned within retries ⇒ FAIL	email / functional	tests/vpn_lan/email_roundtrip.sh (T4.3)	Gated — reachable bridge up creds + a successful send
Open-relay guard (mandatory negative)	§4.3 MANDATORY negative test — an UNAUTHENTICATED relay to an EXTERNAL domain MUST be REFUSED: captured 4xx/5xx of the external RCPT ⇒ PASS; a captured 2xx acceptance ⇒ decisive FAIL (helix_proxy is NEVER an open relay / spam conduit); incomplete probe ⇒	email / security	tests/vpn_lan/email_roundtrip.sh (T4.4)	Gated — reachable bridge up mail server present

Component	Feature	Category	Implementation	Wiring (reachabl
	honest SKIP (never fake-PASS the guard)			

E. Discovery reflector — AUTHORED design + test, remote deploy operator-gated

Component	Feature	Category	Implementation	Wiring (reachable?)
mDNS / DNS-SD / SSDP / WS-Discovery reflector	Remote-side multicast discovery reflector (Avahi enable-reflector for mDNS 5353 / DNS-SD; SSDP relay for 1900; WS-Disc 3702) — multicast is subnet-local and NOT forwarded across the L3 VPN, so a remote-side reflector is required. Client test: avahi-browse -rpt surfaces a resolved (^=) remote service (SCORED); SSDP M-SEARCH count is supplementary/non-scored	discovery / functional	design docs/design/vpn_lan_access/reflector_design.md; test tests/vpn_lan/discovery_reflect.sh	Gated — enumeration reachable once bridge up + re-deployed (HELIX_VPN_REF remote deploy operator-gated (§11.4.122))

F. Casting (Google Cast / DIAL) — AUTHORED, live round-trip operator-gated

Component	Feature	Category	Implementation	Wiring (reach)
Chromecast eureka_info control	Routed GET <code>http://<ip>:8008/setup/eureka_info</code> over the L3 VPN — a real JSON name field is the device-identity evidence; HTTP 200 without a JSON name, or a non-200 answer ⇒ FAIL (fail-closed §11.4.68) Cast status <i>transition</i> observed across two reads (advancing state, NOT a single frozen frame) via an operator-supplied status command; two DISTINCT reads ⇒ transition PASS; identical reads (device idle) ⇒ honest operator_attended SKIP (a transition needs media playing), never a fake PASS	casting / functional	tests/vpn_lan/chromecast_dial.sh (T6.2)	Gated — reach bridge up + HELIX_VPN_CAST control routes TCP
CASTV2 liveness (§11.4.107)	DIAL discovery via the Phase-5 reflector (SSDP M-SEARCH) + <code>_googlecast._tcp</code> mDNS; HTTP control routes over the VPN as	casting / functional (liveness)	tests/vpn_lan/chromecast_dial.sh (T6.3)	Gated — reach bridge up + HELIX_VPN_CAST set
DIAL discovery + control	DIAL discovery via the Phase-5 reflector (SSDP M-SEARCH) + <code>_googlecast._tcp</code> mDNS; HTTP control routes over the VPN as	casting / functional	tests/vpn_lan/chromecast_dial.sh (T6.1 discovery) + reflector_design.md	Gated — disco depends on the (E); control routes bridge up

Component	Feature	Category	Implementation	Wiring (reach)
	ordinary unicast TCP (the routable "cast to a remote display" alternative to Miracast)			

G. Device bridge (ADB over VPN) — AUTHORED; flash operator-blocked

Component	Feature	Category	Implementation	Wiring (reachable?)
ADB connect / shell / getprop	adb connect <host>:5555 over the routed VPN (no proxy hop — recon 4) + central adb-server model; assert OUR serial reaches device state then adb -s <serial> shell getprop ro.product.model returns real content; reachable-but-unusable state (offline/unauthorized) ⇒ FAIL. Safety: touches ONLY the env-configured serial — never kill-server, never a blanket disconnect, never any other/operator/lava-* device	device-bridge / functional	tests/vpn_lan/adb_over_vpn.sh (T7.1/T7.3)	Gated — reachable once bridge up + HELIX_VPN_ADB_H set

Component	Feature	Category	Implementation	Wiring (reachable?)
ADB flash (usbip USB-over-IP)	(§11.4.174); adb disconnect <our-serial> in the cleanup trap Honest boundary (§11.4.6): fastboot is USB-level and network fastboot is USB-bound (recon 4 FACT); the routable path is usbip from a remote host with the device attached. The test NEVER flashes and NEVER invokes fastboot/usbip — it records the boundary and SKIPS operator_attended; any real-device flash is operator-gated (§11.4.133 + §11.4.122)	device-bridge / operator-safety	tests/vpn_lan/adb_over_vpn.sh (T7.4)	Gated — real-device flash requires explicit operator authorization

H. Miracast — Won't-fix, structurally-impossible

Component	Feature	Category	Implementation	Wiring (reachability?)	Re
Miracast over L3 VPN	Won't-fix — structurally-impossible (§11.4.112): Miracast rides Wi-Fi Direct / 802.11 P2P at Layer 2 with no	honest-boundary / structural verdict	docs/design/vpn_lan_access/miracast_verdict.md (cited spec text IS the artifact)	N/A — out of scope by physics; positive capability lives in \$F (Cast/DIAL)	Th do del rec on NE evi the pla con

Component	Feature	Category	Implementation	Wiring (reachable?)	Re
	routable IP hop for an L3 VPN to carry; cited Wi-Fi-Alliance + Wi-Fi-Direct evidence. The routable "cast to a remote display" capability is delivered by Google Cast / DIAL (§F). NO fake traversal test exists or will be authored (a green Miracast-over-VPN result would be fabricated)				cha (\$1 \$1

I. Documentation & verification infrastructure

Component	Feature	Category	Implementation	Y
Operator bridge-setup guide	End-user manual — overview, one-line routing map, the 6-var contract, setup steps, per-protocol matrix, security notes, FAQ; HTML+PDF exports §11.4.65, §11.4.168 visually validated (no	docs / operator	docs/guides/vpn_lan_bridge_setup.md	Y c c e e f

Component	Feature	Category	Implementation	
VPN-LAN Challenge runner	raw diagram source leaking) Runnable anti-bluff harness — runs <code>svord_doctor.sh</code> preflight then each <code>tests/vpn_lan/*.sh</code>, maps exit codes (doctor 0=UP/2=SKIP/3=MISCONFIGURED; tests 0=no-FAIL/1=FAIL), tallies PASS/FAIL/SKIP with per-item logs under host-safety caps (GOMAXPROCS=2 nice ionice, §12.6). Bridge-down ⇒ every item honest SKIP, exit 0 (never a fake PASS); a not-yet-present test ⇒ SKIP, never FAIL	verification / Challenge (§11.4.27)	<code>challenges/scripts/run_vpn_lan_challenges.sh</code>	Y C n i C P - h t
HelixQA <code>vpn_lan.yaml</code> bank	HelixQA test bank — one case per protocol (VLA-BRIDGE-PREFLIGHT / VLA-SMB / VLA-NFS / VLA-FTP / VLA-SFTP / VLA-WEBDAV / VLA-EMAIL / VLA-CAST / VLA-ADB) dispatching <code>ActionTypeShell</code> to the real protocol test so the case verdict IS the real test verdict; documents the exit-code→verdict mapping	verification / HelixQA (§11.4.27)	<code>tools/helixqa/banks/vpn_lan.yaml</code>	C h a h h h v C S

J. Hermetic-tunnel autonomous protocol promotions (§11.4.52) — PROVEN NOW (autonomous, no live VPN, no operator)

What these add. Four protocol legs that were previously *only* operator-gated SKIP now ALSO have an **autonomous PASS path** (all four — Chromecast-eureka, FTP-passive, WebDAV-PROPFIND, Email — plus the H0 substrate are wired into the standing suite as §11.4.135 regression guards via `test_vpn_lan_hermetic`; the substrate + first three landed 71e0bac, email landed b66b2fa closing the §11.4.6 ledger-audit gap): their client-side protocol logic is proven over a **real kernel-WireGuard tunnel** between two unprivileged network namespaces (`unshare -Urnm, veth underlay 10.9.0.x + wg0 overlay 10.10.0.x`), with a **pure-python-stdlib peer** serving the protocol — **zero host installs**, no podman, no live Mullvad, no operator. Each harness drives the **UNMODIFIED** protocol test inside netns A (its `bridge_require` gate flips SKIP→UP), carries a **golden-bad mutation** that MUST make the real assertion FAIL (§11.4.107(10)/§11.4.68 teeth), a **self-fetch not-stale cross-check** (§11.4.107), and a **wrong-destination negative control** (4dd5fc6: a fetch to the UNDERLAY 10.9.0.2 MUST fail — proven load-bearing, `bind-0.0.0.0⇒harness FAILs`; makes §11.4.111 reachability-as-proof self-evidencing, not structural-only), and — on the WG substrate `hermetic_wg_roundtrip.sh` (91af9c6, §11.4.107) — an **underlay-sniff AF_PACKET non-leak differential** (during the round-trip, capture on the underlay `veth0` and assert BOTH WG ciphertext present, type-4 0x04 datagram to :51820, AND the per-run plaintext nonce ABSENT in raw underlay bytes; load-bearing golden-bad `SNIFF_MUT=plain` flips ONLY "plaintext absent" to FAIL 3/3, honest exit-3 on empty capture, honest SNIFF-SKIP when unavailable; independent §11.4.142 review GO 11/11 — fan-out to the other 4 harnesses tracked as a follow-up) — all **3/3 deterministic** (§11.4.50) and **independently reviewed to GO** (§11.4.142; the WG negative control was caught as a tautology in review round 1 and fixed before landing — §11.4.107(10)). **Honest boundary (§11.4.6)**: these prove the *client-side protocol logic* over an encrypted tunnel against a *controlled* peer — they do NOT prove the real live server on the Mullvad topology (that stays the operator-gated live round-trip above), and the peer bodies are controlled fixtures. They COMPLEMENT, never replace, the live rows.

Component	Feature	Category	Implementation
H0 hermetic WG substrate	Rootless two-netns kernel-WireGuard round-trip (<code>veth underlay + real wg0 overlay</code>); the reusable	test-harness / foundation	<code>tests/vpn_lan/hermetic_netns_poc.sh</code> (<code>veth L3</code>) + <code>hermetic_wg_roundtrip.sh</code> (full WG)

Component	Feature	Category	Implementation
	substrate every H2.x promotion builds on		
Chromecast eureka_info — hermetic promotion	The UNMODIFIED chromecast_dial.sh eureka control leg run AUTONOMOUSLY over the hermetic tunnel against a stdlib eureka_info JSON peer on 10.10.0.2:8008; golden-bad = a nameless body ⇒ the real test FAILs fail-closed	casting / functional (autonomous)	tests/vpn_lan/hermetic_bridge_run.sh
FTP passive — hermetic promotion	The UNMODIFIED ftp_sftp_webdav.sh FTP leg run AUTONOMOUSLY over the hermetic tunnel against an embedded ~85-line stdlib FTP server on 10.10.0.2:2121 (PASV advertises 10.10.0.2, EPSV carries none — both traverse wg0); harness ADDITIONALLY content-verifies a byte RETR of a fresh nonce (§11.4.107(9)); golden-bad = empty listing ⇒ the real test SKIPs (no PASS)	file-transfer / functional (autonomous)	tests/vpn_lan/hermetic_ftp_run.sh
WebDAV PROPFIND — hermetic promotion	The UNMODIFIED ftp_sftp_webdav.sh WebDAV leg run AUTONOMOUSLY over the hermetic tunnel: a stdlib 207 Multi-Status origin on 10.10.0.2:8080 reached THROUGH a stdlib forward proxy on	file-transfer / functional (autonomous)	tests/vpn_lan/hermetic_webdav_run.sh

Component	Feature	Category	Implementation
Email round-trip (SMTPS/ IMAPS/ POP3S) — hermetic promotion	127.0.0.1:3128 (RFC 7230 §5.3.2→§5.3.1); golden-bad = origin flips to 200 ⇒ the real test FAILs fail-closed (§11.4.68) — the mutation surface IS exactly the promoted test's only PASS gate The UNMODIFIED email_roundtrip.sh run AUTONOMOUSLY over the hermetic tunnel against a pure-stdlib implicit-TLS mail peer (SMTPS 465 / IMAPS 993 / POP3S 995, shared in-memory mailbox) on 10.10.0.2; SMTP DATA embeds a per-run nonce that POP3S RETR MUST return (round-trip oracle §11.4.107); two golden-bad teeth — MAIL_MUT=openrelay ⇒ real FAIL: open_relay_refused (external RCPT accepted), MAIL_MUT=droptoken ⇒ real FAIL: pop3s_retrieve_roundtrip (token not retrieved) — each fails ONLY its target leg	email / functional (autonomous)	tests/vpn_lan/ hermetic_email_run.sh

Not hermetic-promotable — honest verdict (§11.4.6 / §11.4.3). The clean zero-install promotion set is now **COMPLETE** (Cast-eureka + FTP + WebDAV + email above). The remaining protocol legs are genuinely operator-gated: SFTP needs an SSH server (sshd absent on host + no stdlib SSH server; paramiko/asyncssh are pip → §11.4.122); SMB/NFS need samba/nfsd (absent + root); the discovery_reflect.sh scored mDNS leg hard-requires the avahi-browse *client* (absent) so a stdlib mDNS responder wouldn't help the UNMODIFIED test; ADB/container need a device/podman. **Hardening landed:** the wrong-destination negative control across all 5 harnesses (4dd5fc6, §11.4.111 now self-evidencing) + the /usr/sbin/wg preflight (already satisfied in

all 5, verified §11.4.6) + the **underlay-sniff AF_PACKET non-leak differential on the WG substrate** (91af9c6, §11.4.107, independent §11.4.142 GO 11/11). **Hardening landed (cont.):** the Ethernet-ethertype guard on the sniff frame analyzer LANDED (cdb0ccd, task #64, correct-by-construction + a load-bearing crafted-frame self-test proven by guard-removal; independent §11.4.142 GO 5/5). **Next (tracked):** fan the sniff differential out to the protocol harnesses (task #65) — the design pass (§11.4.150) found it is meaningful for **3** of them (bridge/ftp/webdav, all plaintext-under-WG) but **N/A for email** (implicit-TLS encrypts the token *below* WG, so "plaintext absent on the underlay" is tautologically true regardless of the tunnel — building it there would be a §11.4-forbidden vacuous test; honest documented N/A instead); the email leg is now wired into the standing suite (b66b2fa, task #66 done — the §11.4.6 ledger-audit gap closed, independent review a5bc6266 GO, all 5 rows PASS). Deep-research: [../../research/hermetic_protocol_promotion_20260702/FINDINGS.md](https://github.com/0x00sec/hermetic_protocol_promotion_20260702/FINDINGS.md) (Rev 5, §6 mail + §7 underlay-sniff IMPLEMENTED + §7.1 fan-out design).

§11.4.169 test-type coverage note (honest boundary)

The autonomously-provable layers today are: **unit / config** (bridge-contract resolution + set-u safety), the **standing-suite gate** (test_vpn_lan_bridge — 3-verdict discrimination + §11.4.115 UP-stub teeth, GREEN), the **Challenge runner** (runnable NOW, honest bridge-down all-SKIP tally), and the **structural verdict** (Miracast, cited). The **integration / e2e / full-automation / security / stress+chaos** live layers per protocol are AUTHORED with evidence-gating + wrong-answer-FAIL teeth but **honestly SKIP** until the operator connects the sword bridge (§11.4.3) — they are never counted as PASS on a down bridge. HelixQA autonomous-session coverage is **run-blocked** (6 un-vendored siblings, §11.4.3), surrogated by the Challenge runner. No metadata-only / config-only / absence-of-error PASS is recorded anywhere in this ledger (§11.4 / §11.4.1 / §11.4.69).

Honest boundary (§11.4.6)

What is PROVEN NOW, autonomously, with committed evidence: the decoupled env-var bridge contract, the sword-doctor 3-verdict preflight with §11.4.115 teeth in the GREEN standing suite, the honest-SKIP anti-bluff gate across every protocol test (bridge-down ⇒ SKIP + exit 0 + zero ^PASS: lines — never a fake PASS), the wrong-answer-FAIL teeth (WebDAV non-207, SFTP/NFS/SMB sha-mismatch, open-relay external-RCPT-accepted, ADB

not-device-state, Cast non-200/no-name), the runnable Challenge harness, the operator guide, and the Miracast structural-impossibility verdict. What is **operator-gated** and therefore SKIP/OPERATOR-BLOCKED until unblocked: every **live** protocol round-trip (real bytes, sha256, mailbox content, eureka JSON, device serials — needs the sword bridge up), the multicast-reflector remote deployment (§11.4.122), any real-device flash (§11.4.133), and the autonomous HelixQA session (6 un-vendored siblings). No live capability is marked PASS without captured evidence on a genuinely-up bridge, and no feature is "done" for its live capability until its runtime signature verifies on a live bridge (§11.4.108) and it crosses the §11.4.169 test-type matrix + independent review (§11.4.142). This ledger does not substitute for the §11.4.40 full-suite retest before any release tag.

Feature tally: 4 rows PASS-now foundation (bridge contract · sword-doctor preflight · operator guide · Challenge runner) · **5 rows PASS-now §J hermetic autonomous protocol promotions** (H0 WG substrate · Chromecast-eureka · FTP-passive · WebDAV-PROPFIND · Email — client-side protocol logic proven over a hermetic tunnel, zero installs, golden-bad teeth, 3/3 deterministic, independent GO each). **Standing-suite wiring (§11.4.135 guards via test_vpn_lan_hermetic):** all 5 are wired — the substrate + Cast/FTP/WebDAV (71e0bac) and now the **email leg (b66b2fa, task #66)**, closing the §11.4.6 gap the §11.4.169 ledger audit caught (independent review a5bc6266 GO, all 5 rows PASS in the real loop). · 16 rows operator-gated SKIP/OPERATOR-BLOCKED for their **live Mullvad** round-trip (all live protocol round-trips + reflector deploy + ADB flash + HelixQA run) · 1 row Won't-fix (Miracast, structurally-impossible §11.4.112).